

$$= -3.2 \times 10^{-6} \text{ C [A1]}$$

- 7 (a) (i) The power bank converts 12.0 J of electrical energy from chemical energy (or other forms) [B1]

when one coulomb of charge passes through [B1]

(ii) Lamps A, B and C will be turned on with equal brightness [B1]

(iii) Note that all the lamps and devices are parallel to each other.

Total effective resistance

$$= 3.0 + \left(\frac{1}{25.0} + \frac{1}{25.0} + \frac{1}{25.0} + \frac{1}{25.0} + \frac{1}{25.0} + \frac{1}{38.0} + \frac{1}{42.0} \right)^{-1} \quad [\text{C1}]$$

$$= 6.998 \, \Omega$$

$$I = \frac{V}{R} = \frac{12.0}{6.998} = 1.71 \text{ A} \quad [\text{A1}]$$

(iv)

$$V_{\text{terminal}} = E - Ir$$

$$= 12.0 - (1.71)(3.0) \quad [\text{C1}]$$

$$= 6.86 \text{ V} \quad [\text{A1}]$$

- (v) Effective resistance across the lamps increase, hence p.d. across lamps increase (by potential divider principle). [B1]

Since $P = \frac{V^2}{R}$, power dissipated by lamps increase, hence brightness increase. [B1]

OR

Total resistance across the terminals of the power bank has increased, so current through the powerbank has decreased, so terminal p.d. has increased

$$(V_{\text{terminal}} = E - Ir).$$

Since $P = \frac{V_{\text{terminal}}^2}{R}$, power dissipated by lamps increase, hence brightness increase.

(b) (i)

$$V_{XY} = \frac{7.2}{7.2 + 2.0}(18.0) = 14.087 \text{ V} \quad [\text{C1}]$$

$$V_{\text{power bank}} = \frac{25.0}{25.0 + 3.0}(12.0) = 10.714 \text{ V} \quad [\text{M1}]$$

$$V_{\text{power bank}} = \frac{L}{L_{XY}} V_{XY}$$

$$10.714 = \frac{L}{1.00}(14.087)$$

$$L = 0.761 \text{ m} \quad [\text{A1}]$$

(ii) To reduce percentage or fractional uncertainty of balance length [B1]

(iii) XY will have higher resistance, thus higher p.d. across XY by potential divider principle. [M1]

Thus a smaller balance length is needed to balance p.d. across the power bank.
[A1]